Decodelabs Data Analytics Project 1

Data Cleaning

Objective

The main objective of the project is to clean and transform a raw e-commerce dataset so that it becomes ready for Analisis

Specifically, the project aims to:

a. Identify and handle missing values.

b. Find and remove duplicates records.

c. Identify and correct incorrect data formats, expecially dates, numbers and text.

d. Ensure the dataset has zero duplicates.

e. Ensure dates are correctly formated.

Import libraries

Load dataset

In [95]:
```
import pandas as pd
import numpy as np
df = pd.read_excel("Dataset for Data Analytics.xlsx")
df.head()
```

Out[95]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SA
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SA
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREI
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SA
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SA

Check the Shape of the Dataset

In [27]:

```
print("Dataset Shape")
df.shape
```

```
Dataset Shape
```

Out[27]:

```
(1200, 14)
```

Print All Columss Name

In [29]:

```
print("\nColumn Names:")
print(df.columns)
```

```
Column Names:
Index(['OrderID', 'Date', 'CustomerID', 'Product', 'Quantity', 'UnitPrice',
       'ShippingAddress', 'PaymentMethod', 'OrderStatus', 'TrackingNumber',
       'ItemsInCart', 'CouponCode', 'ReferralSource', 'TotalPrice'],
      dtype='object')
```

Print Dataset Information

In [32]:

```
print("\nDataset Information:")
df.info()
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   OrderID          1200 non-null   object
 1   Date             1200 non-null   datetime64[ns]
 2   CustomerID       1200 non-null   object
 3   Product          1200 non-null   object
 4   Quantity         1200 non-null   int64
 5   UnitPrice        1200 non-null   float64
 6   ShippingAddress  1200 non-null   object
 7   PaymentMethod    1200 non-null   object
 8   OrderStatus      1200 non-null   object
 9   TrackingNumber   1200 non-null   object
 10  ItemsInCart      1200 non-null   int64
 11  CouponCode       891 non-null    object
 12  ReferralSource   1200 non-null   object
 13  TotalPrice       1200 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(9)
memory usage: 131.4+ KB
```

Check Basic Statistics

In [34]:

```python
df.describe()
```

Out[34]:

	Date	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200	1200.000000	1200.000000	1200.000000	1200.000000
mean	2024-03-22 16:58:48	2.945833	356.412750	5.485000	1053.968300
min	2023-01-01 00:00:00	1.000000	11.390000	1.000000	11.390000
25%	2023-08-03 18:00:00	2.000000	186.062500	4.000000	410.520000
50%	2024-03-23 00:00:00	3.000000	364.210000	5.000000	823.615000
75%	2024-11-08 12:00:00	4.000000	521.570000	7.000000	1578.475000
max	2025-06-30 00:00:00	5.000000	699.930000	10.000000	3456.400000
std	NaN	1.407557	197.177146	2.281983	819.856558

Check for Missing Value

About 309 missig values found on the coupon code column

In [37]:
```
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
OrderID              0
Date                 0
CustomerID           0
Product              0
Quantity             0
UnitPrice            0
ShippingAddress      0
PaymentMethod        0
OrderStatus          0
TrackingNumber       0
ItemsInCart          0
CouponCode         309
ReferralSource       0
TotalPrice           0
dtype: int64
```

In [39]:
```
missing_percent = (df.isnull().sum() / len(df)) * 100
missing_report = pd.DataFrame({
"Missing Values": df.isnull().sum(),
 "Missing Percentage": missing_percent
})
missing_report
```

Out[39]:

	Missing Values	Missing Percentage
OrderID	0	0.00
Date	0	0.00
CustomerID	0	0.00
Product	0	0.00
Quantity	0	0.00
UnitPrice	0	0.00
ShippingAddress	0	0.00
PaymentMethod	0	0.00
OrderStatus	0	0.00
TrackingNumber	0	0.00
ItemsInCart	0	0.00
CouponCode	309	25.75
ReferralSource	0	0.00
TotalPrice	0	0.00

Missing Values was replaced with NoCode

In [42]:

```
df["CouponCode"]= df["CouponCode"].fillna("NoCode")
print("\nMissing Values after Cleaning:")
print(df.isnull().sum())
```

```
Missing Values after Cleaning:
OrderID            0
Date               0
CustomerID         0
Product            0
Quantity           0
UnitPrice          0
ShippingAddress    0
PaymentMethod      0
OrderStatus        0
TrackingNumber     0
ItemsInCart        0
CouponCode         0
ReferralSource     0
TotalPrice         0
dtype: int64
```

In [44]:
```python
df["CouponCode"].head(20)
```

Out[44]:
```
0       SAVE10
1       SAVE10
2     FREESHIP
3       SAVE10
4       SAVE10
5       SAVE10
6       SAVE10
7     FREESHIP
8       NoCode
9       SAVE10
10    WINTER15
11      SAVE10
12    FREESHIP
13    FREESHIP
14      SAVE10
15      NoCode
16    FREESHIP
17      NoCode
18      SAVE10
19      SAVE10
Name: CouponCode, dtype: object
```

No duplicates rows

In [47]:
```python
print("\nduplicate rows:")
print(df.duplicated().sum())
```

```
duplicate rows:
0
```

In [49]:
```
print(df["OrderID"].duplicated().sum())
print("\nduplicate rows:")
print(df.duplicated().sum())
```

```
0

duplicate rows:
0
```

In [53]:
```
df = df.drop_duplicates()
df = df.drop_duplicates(subset = "OrderID")
df
```

Out[53]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Cou
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	
...	...	...	...	...	...	...	...	...	...	...	...	
1195	ORD201195	2024-06-20	C21126	Desk	1	107.04	392 Main St	Credit Card	Cancelled	TRK38009181	6	
1196	ORD201196	2024-03-04	C20095	Monitor	2	662.53	778 Main St	Online	Cancelled	TRK69207593	5	
1197	ORD201197	2023-07-13	C79674	Tablet	2	436.84	275 Main St	Online	Delivered	TRK88039356	2	
1198	ORD201198	2024-08-22	C64753	Chair	4	262.52	509 Main St	Debit Card	Cancelled	TRK71683331	4	V
1199	ORD201199	2023-06-11	C57502	Tablet	4	560.58	201 Main St	Gift Card	Returned	TRK51116746	6	

1200 rows × 14 columns

In [55]:
```
print(df.duplicated().sum())
print(df["OrderID"].duplicated().sum())
```

```
0
0
```

Clean white space

In [58]:
```
df["Product"] = df["Product"].str.strip().str.title()
df.head(2)
```

Out[58]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	S/
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	S/

In [60]:
```
df["ShippingAddresst"] = df["ShippingAddress"].str.strip().str.title()
df.head(2)
```

Out[60]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	S/
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	S/

Round Numeric column to 2 decimalplace

In [63]:
```
df["UnitPrice"] = df["UnitPrice"].round(2)
df.head(2)
```

Out[63]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	S/
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	S/

In [65]:
```
df["TotalPrice"] = df["TotalPrice"].round(2)
df.head(2)
```

Out[65]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SA
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SA

The Clean dataset wa exported and saved as csv

In [68]:

```
df.to_csv("Cleaned_dataset.csv", index = False)
df.head()
```

Out[68]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SA
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SA
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FRE
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SA
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SA

In [70]:

```
print("Dataset saved successfully!")
```

```
Dataset saved successfully!
```

In [74]:

```
df.head(10)
```

Out[74]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Coupon
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SA
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SA
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FRE
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SA
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SA
5	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SA
6	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SA
7	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FRE
8	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	N
9	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SA

Conclussion

The raw dataset was successfully cleaned and transformed making it ready and fit for analysis with no missing values and no duplicates rows

In [ ]: